TEROS12 Testbench Setup

The software setup is intended for Windows. If you are on a Mac please let me know so I can fix the guide

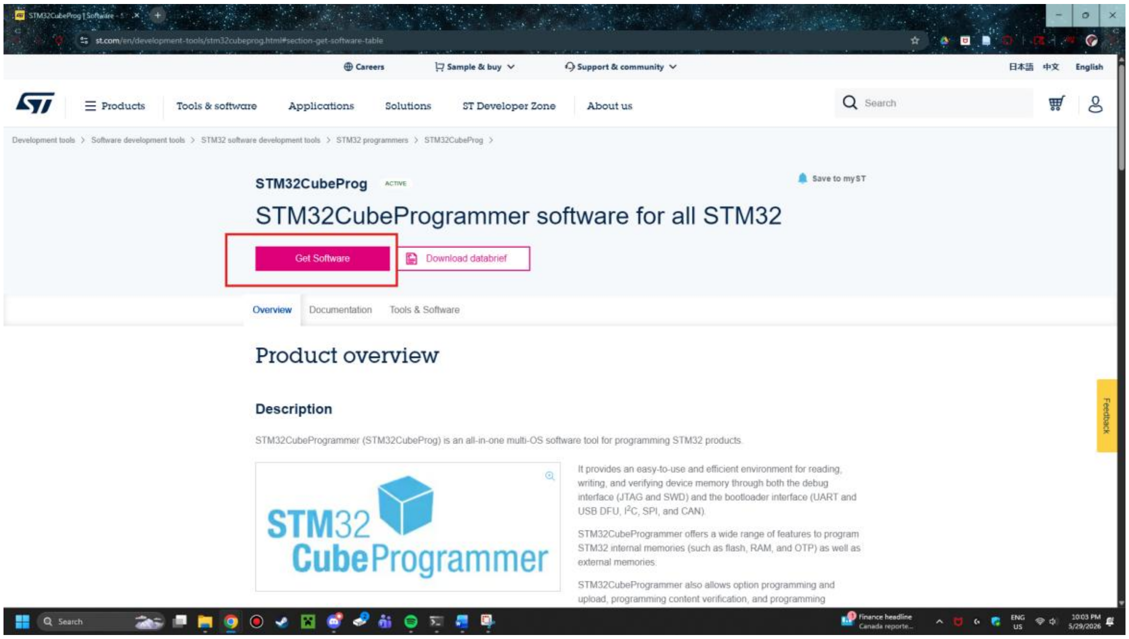
Software Downloads:

Arduino IDE: https://www.arduino.cc/en/software/

- Select your operating system and launch the executable file downloaded

STM32CubeProgrammer: https://www.st.com/en/development-tools/stm32cubeprog.html#section-get-software-table

- Go to “Get Software”



- Select your operating System (Win64) and click “Get Latest”

Get Software

	Part Number	General Description	Latest version	Supplier	Download	All versions
+	CubePrg-Mac-Arm	STM32CubeProgrammer software for MacOS Arm	2.22.0	ST	Get latest	Select version
+	STM32CubePrg-Lin	STM32CubeProgrammer software for Linux	2.22.0	ST	Get latest	Select version
+	STM32CubePrg-Mac	STM32CubeProgrammer software for Mac	2.22.0	ST	Get latest	Select version
+	STM32CubePrg-W32	STM32CubeProgrammer software for Win32	2.22.0	ST	Get latest	Select version
+	STM32CubePrg-W64	STM32CubeProgrammer software for Win64	2.22.0	ST	Get latest	Select version

GET-SOFTWARE-RECOMMENDS

- Accept the license terms
- When prompted, click “Download as a guest”

Thanks for having accepted the license.
Please log in to MyST or create an account to start the download.

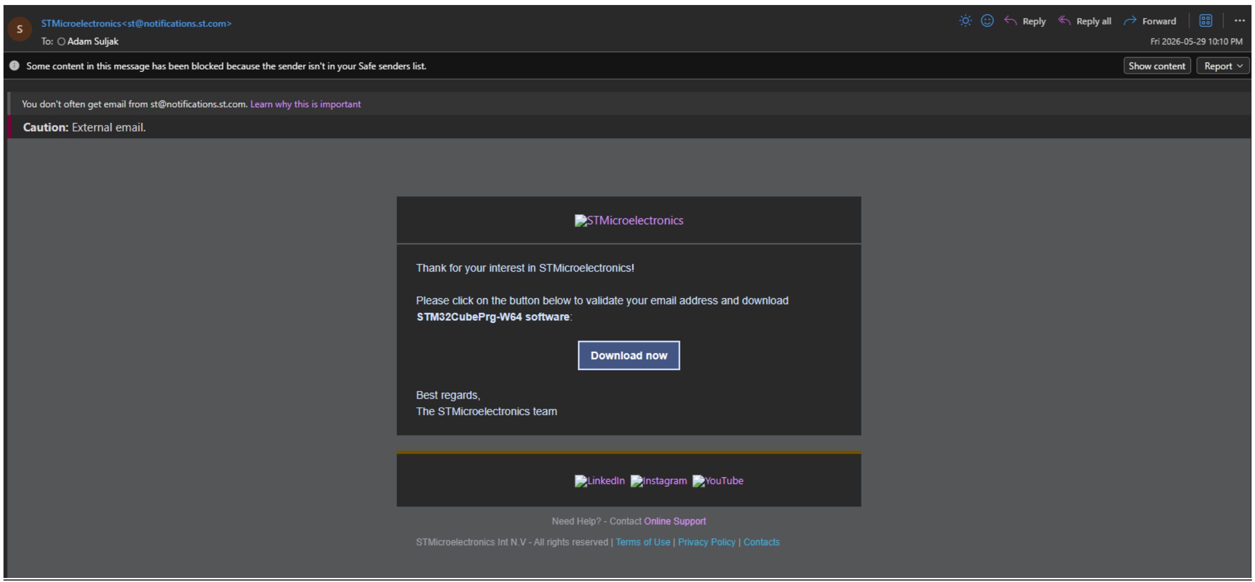
Log in to MyST

Create a MyST account in less than 1 minute

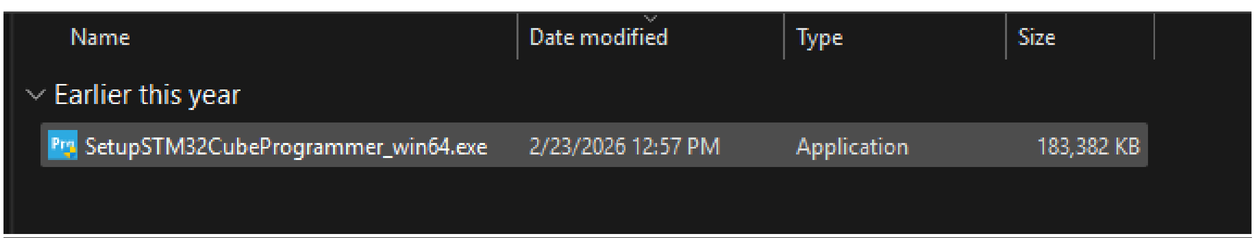
Download as a guest

- Enter your name and email (I used my McMaster email for this). Do NOT click to get future updates about the software. Then click “Get Link to Download”

- Open your email where you should have a link to the installer. Click “Download Now”

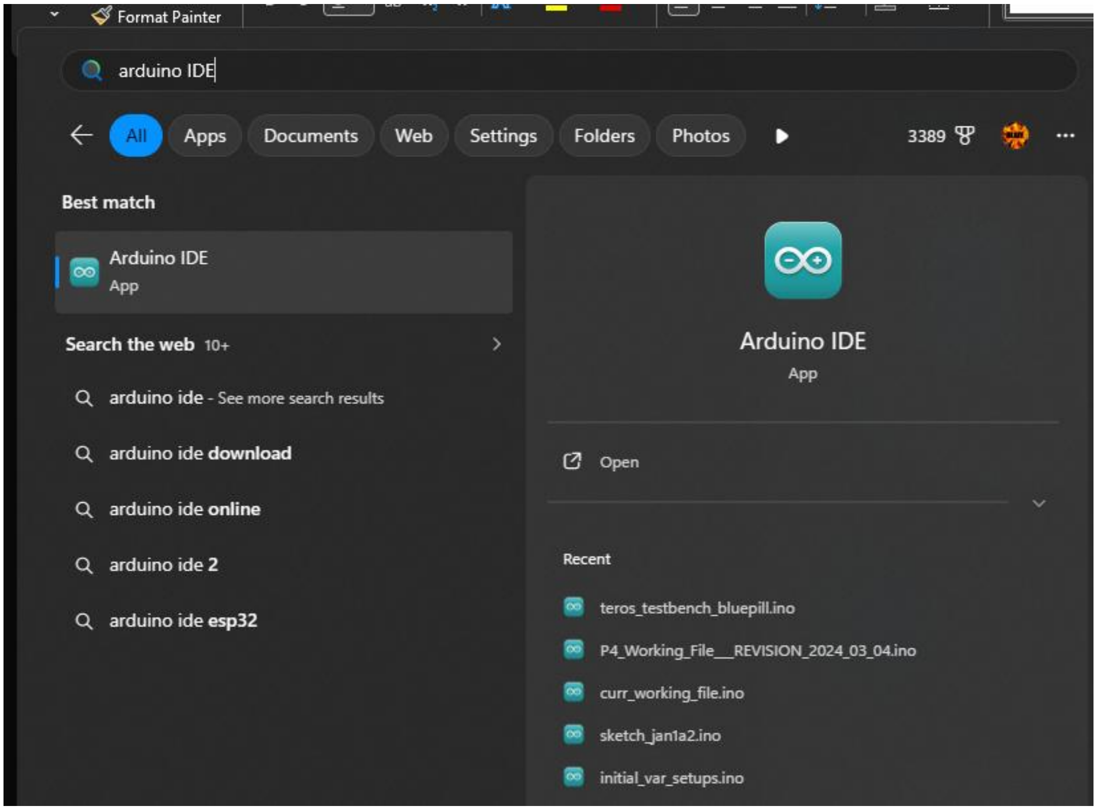


- Unzip the file (Right Click > Extract All).
- In the unzipped folder, run the installer inside and follow all the default setup

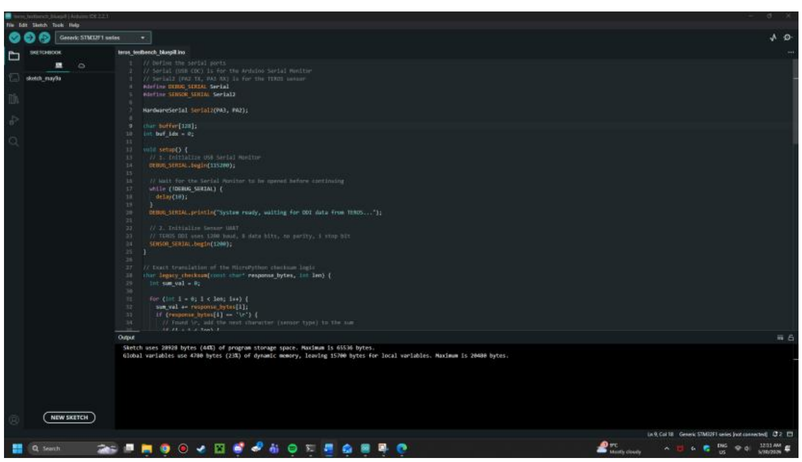


STM32Duino Configuration

- Open Arduino IDE

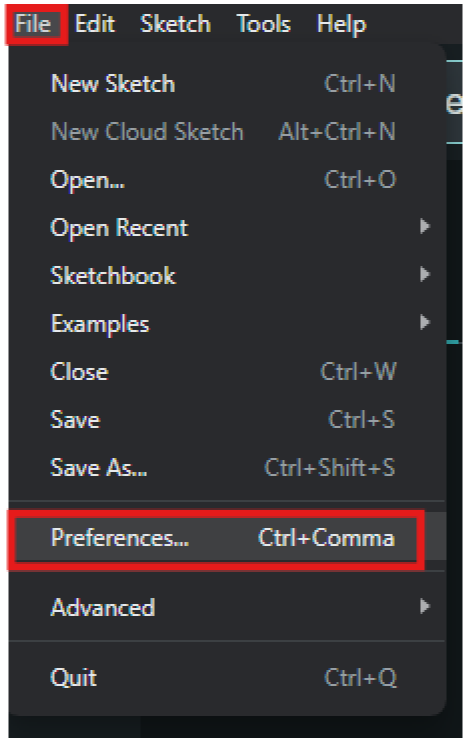


- Once opened, a blank file will open. Click on File > Open and navigate to the directory where you downloaded the TEROS12 code. **Select teros_testbench_bluepill.ino and click open**
- The file should then open, and look something like this

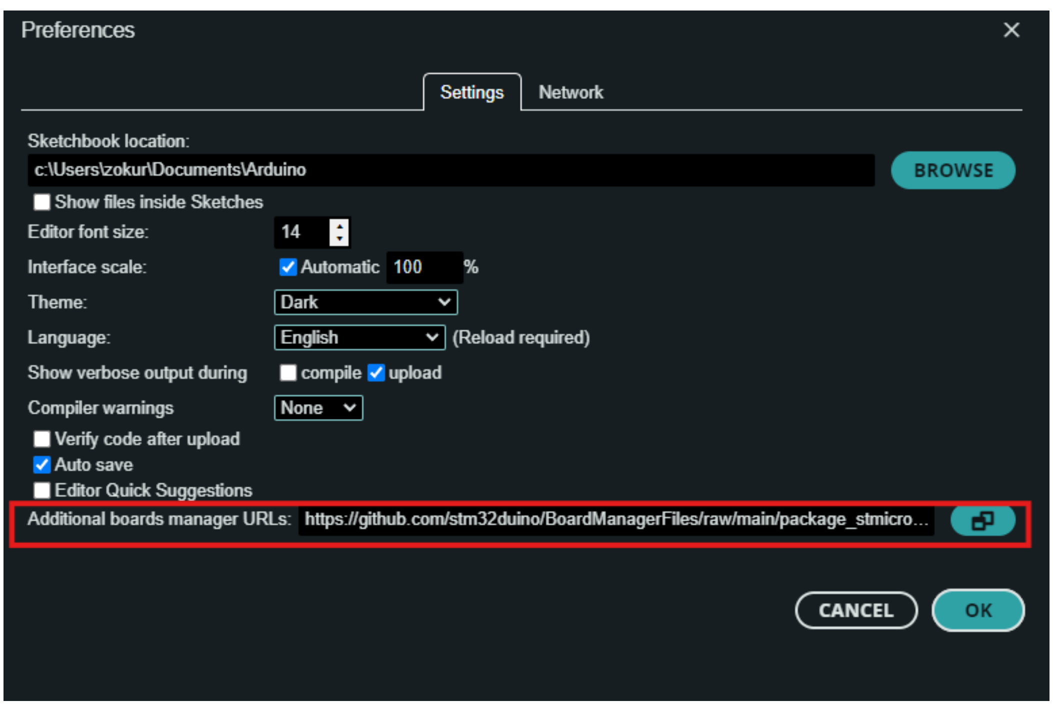


The following steps will walk through how to configure the STM32Duino environment.

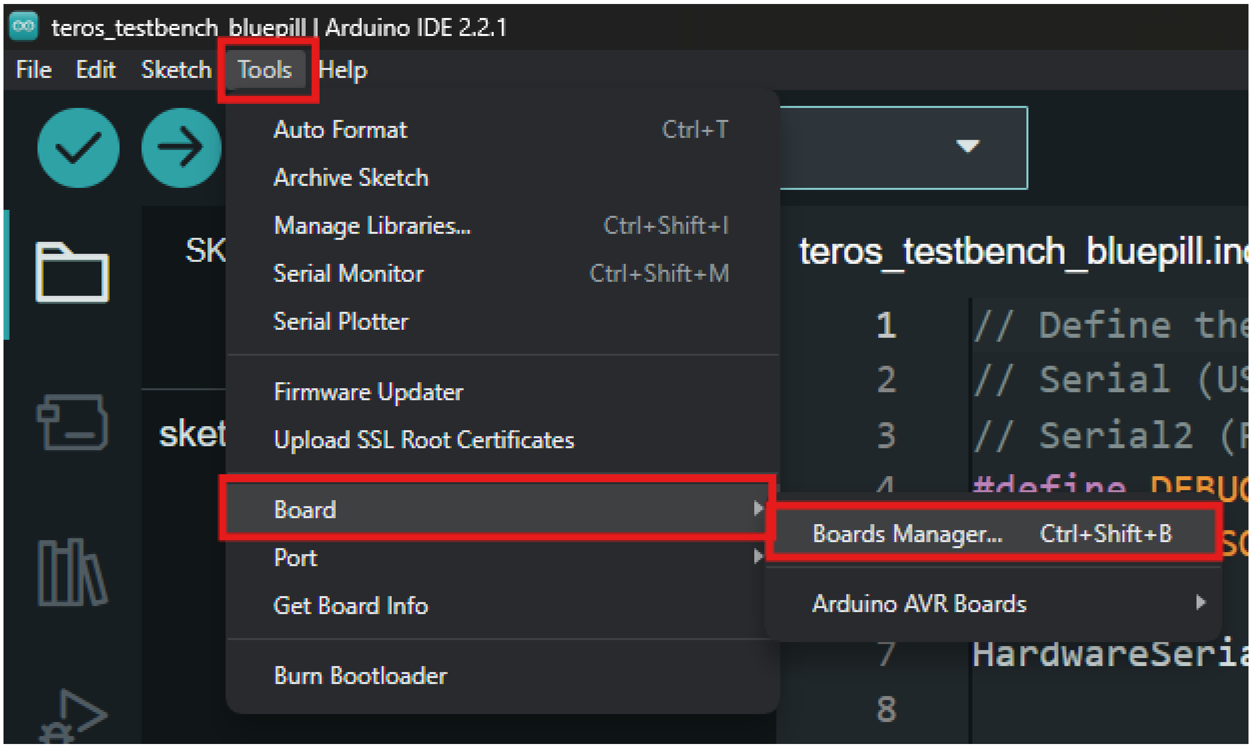
- Go to File > Preferences



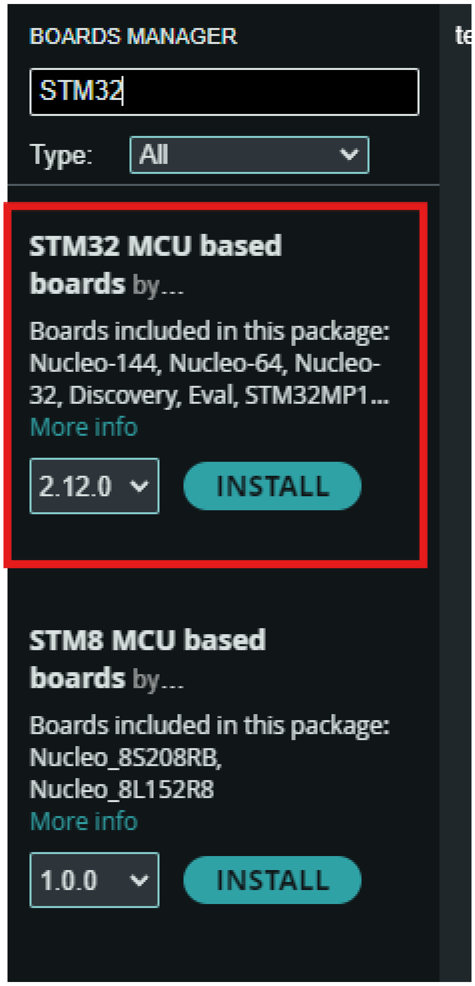
- In the “Additional Boards Manager URLs field” paste the following URL: https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json. Following this, click OK.



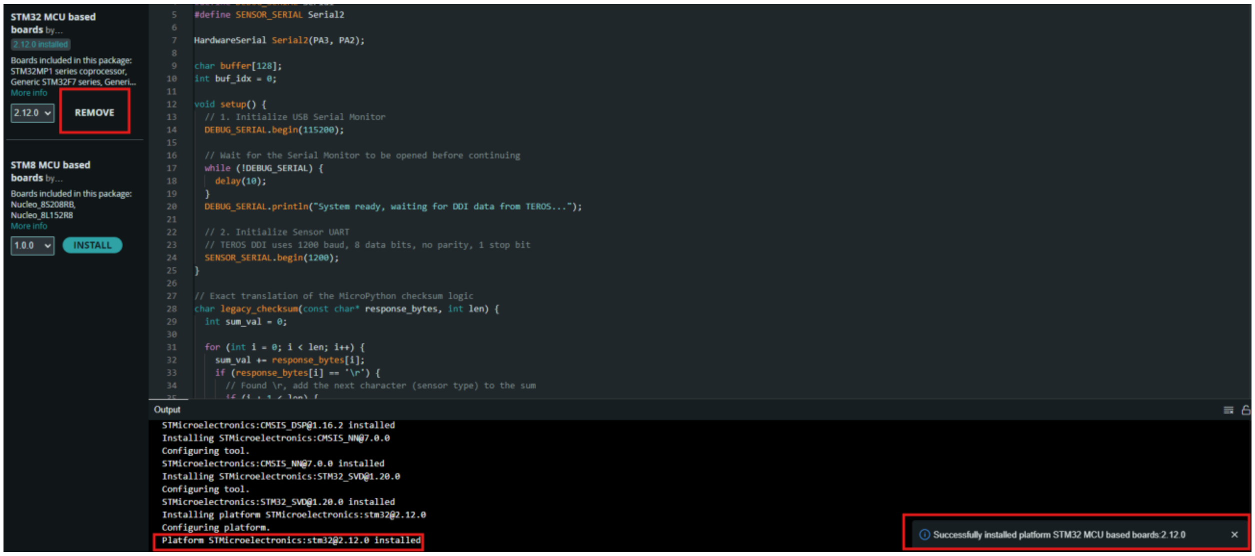
- Next, click on Tools > Board > Boards Manager



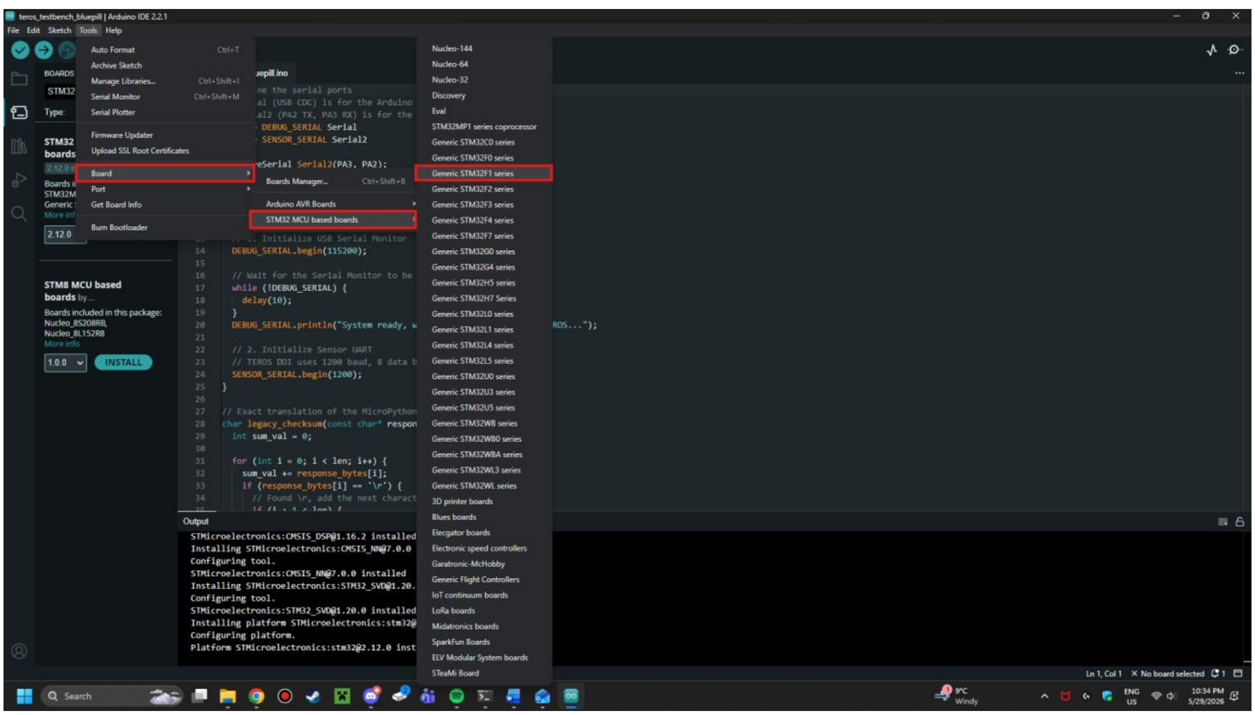
- Search for STM32, find the package named STM32MCU and install it (install the latest version. Make sure it says STM32, and **not** STM8. Once this is done, let the install run



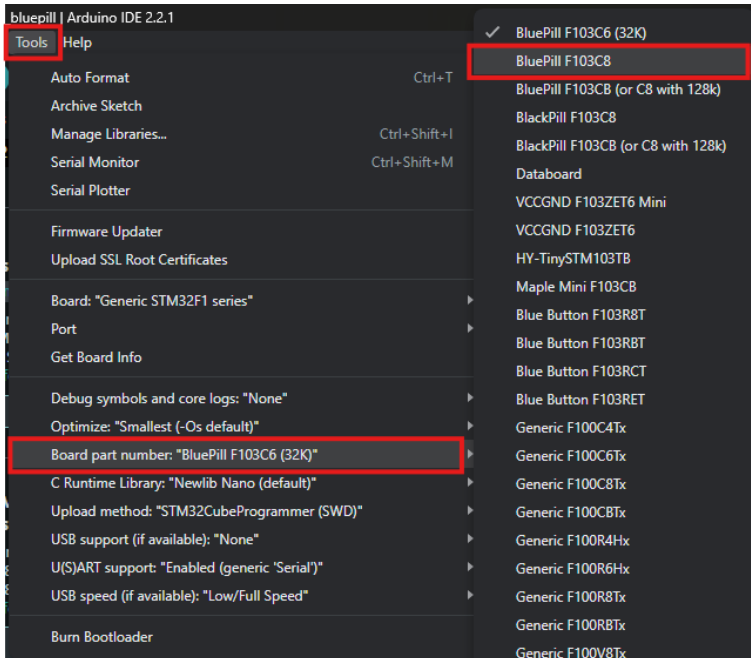
- The installation will be finished when you see the following messages, as well as the “Remove” button beside “STM32 MCU based boards” (do not hit the remove button)



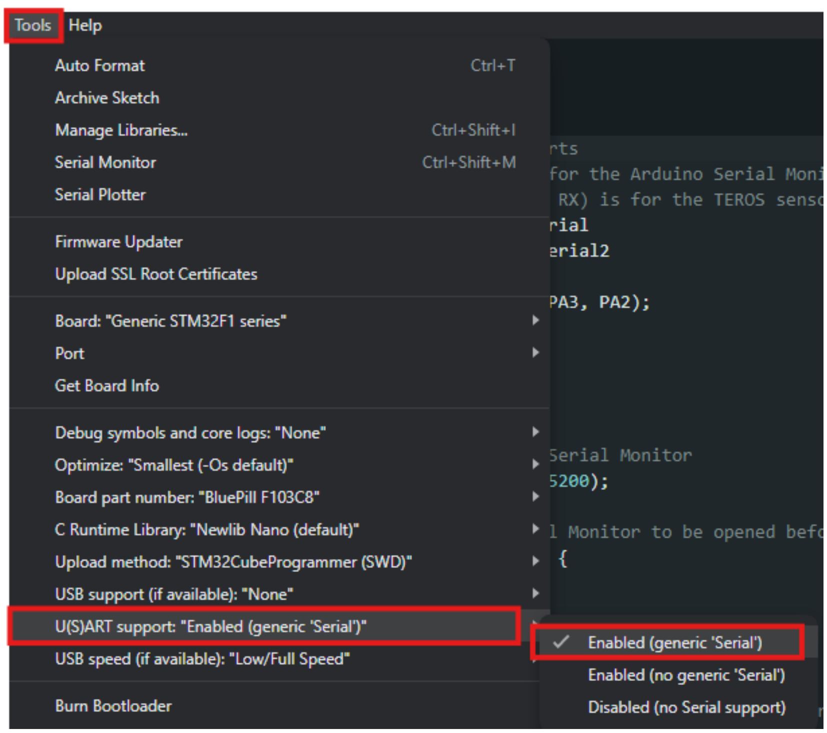
- Once installation is done, go to Tools > Board > STM32 MCU Based Boards
  - Select Generic STM32F1



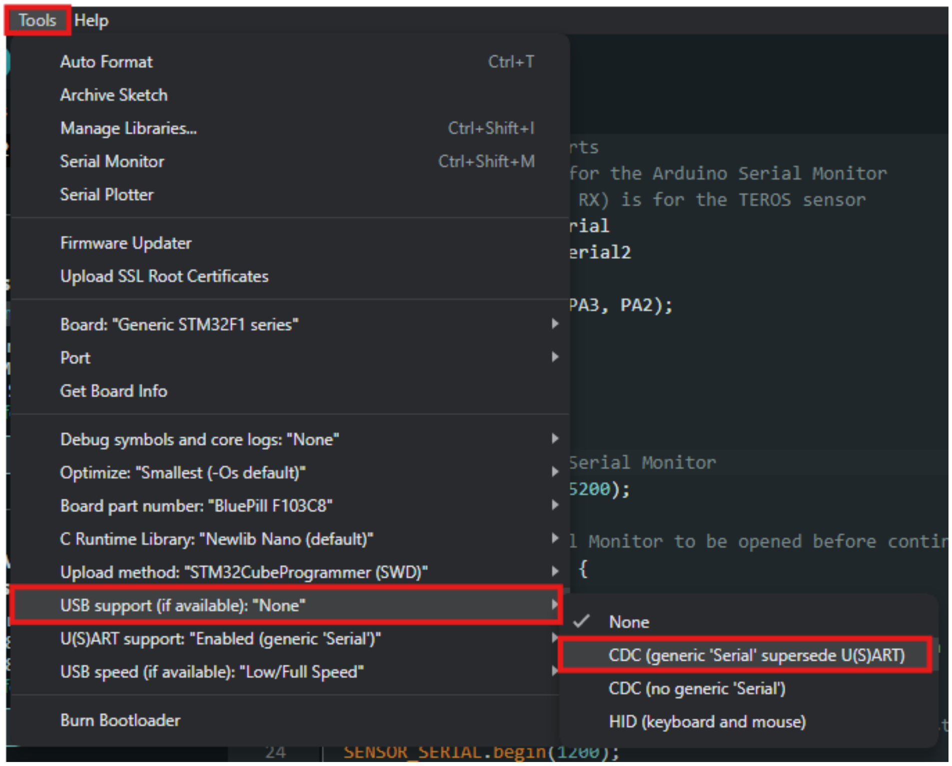
- In the tools menu, ensure the following settings are met. The settings may need to be changed using the subsequent option menus that appear.
  - Board Part Number: BluiePill F103C8



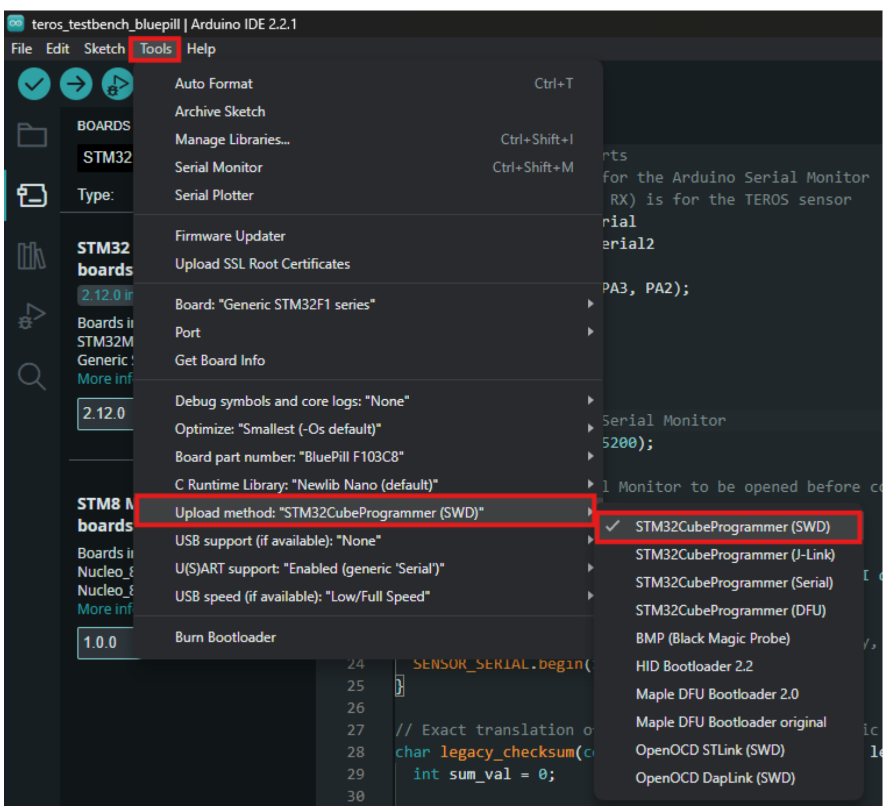
  - U(S)ART Support: Enabled (generic “serial”)”



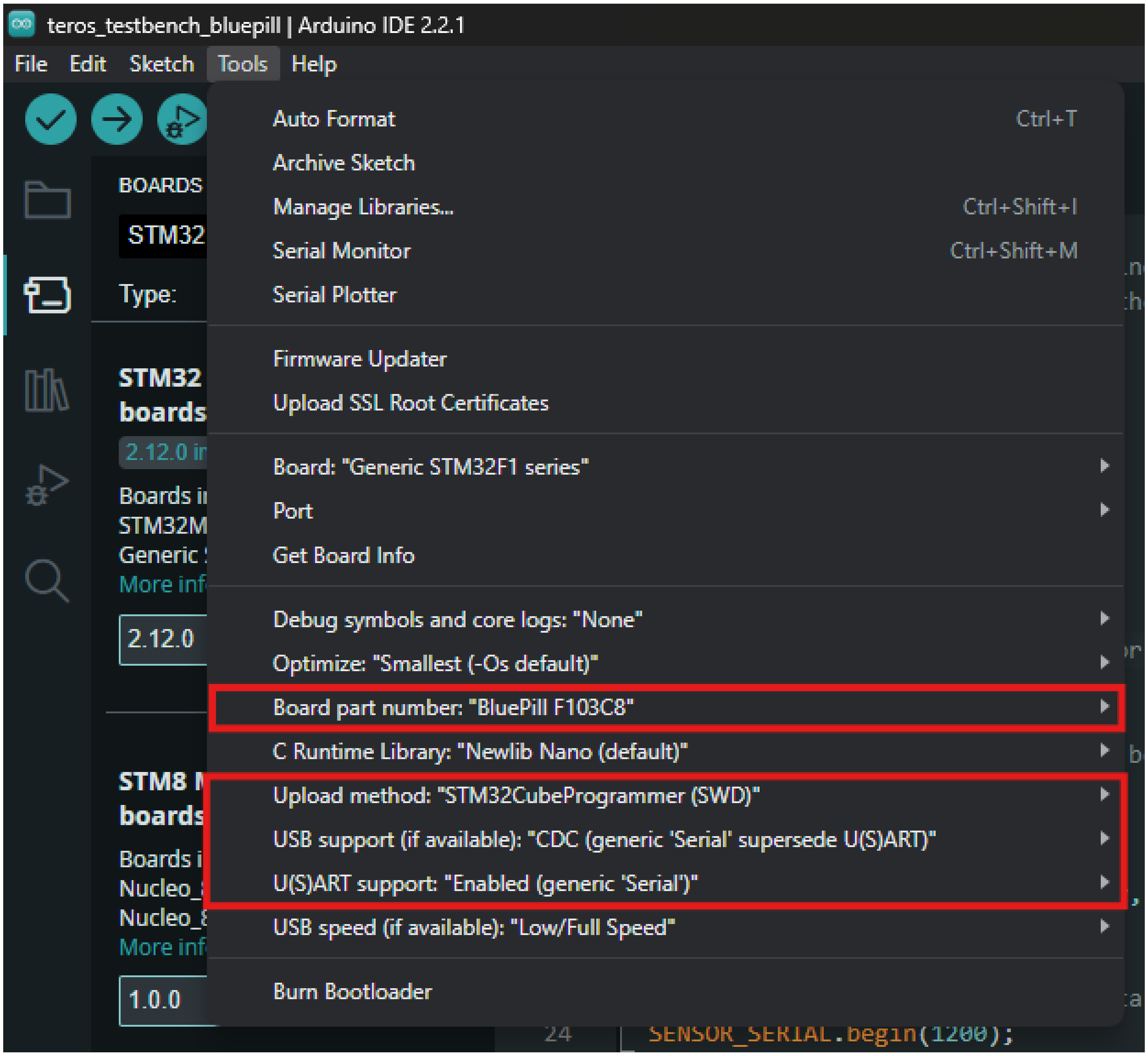
- USB Support (if available): CDC (generic 'Serial' supersede U(S)ART)"



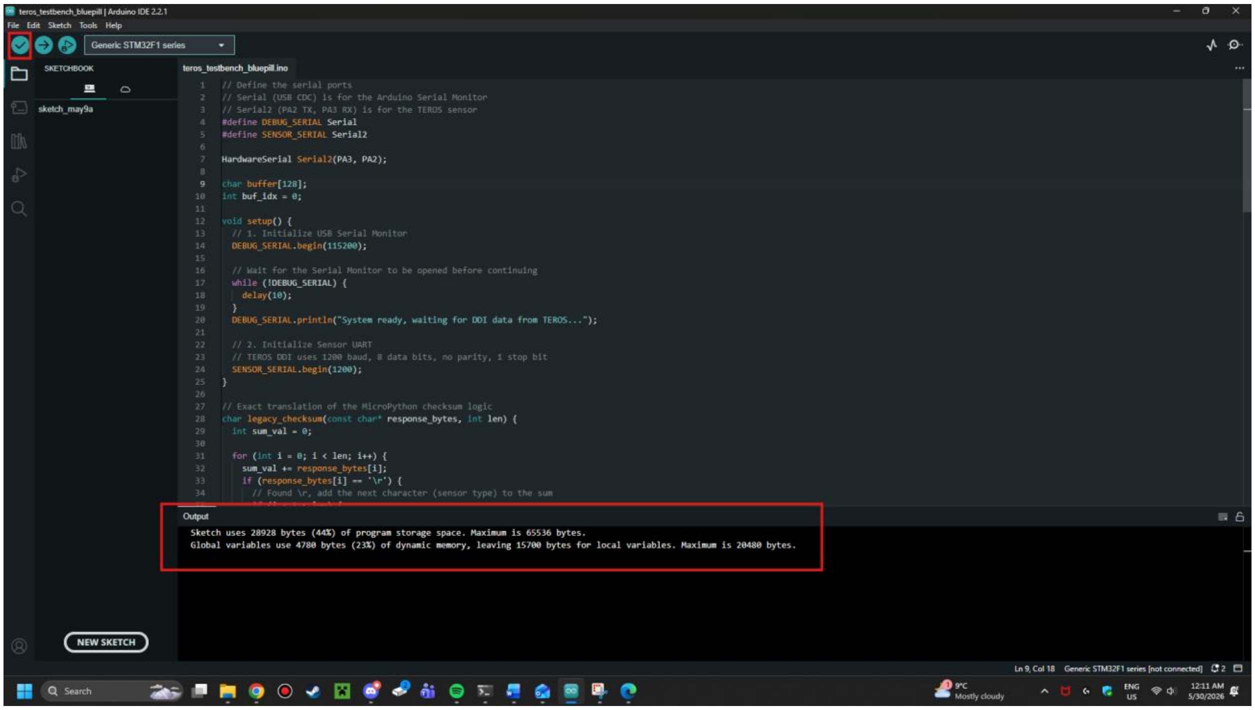
- Upload method: STM32CubeProgrammer(SWD)'



- When complete, it should look like this



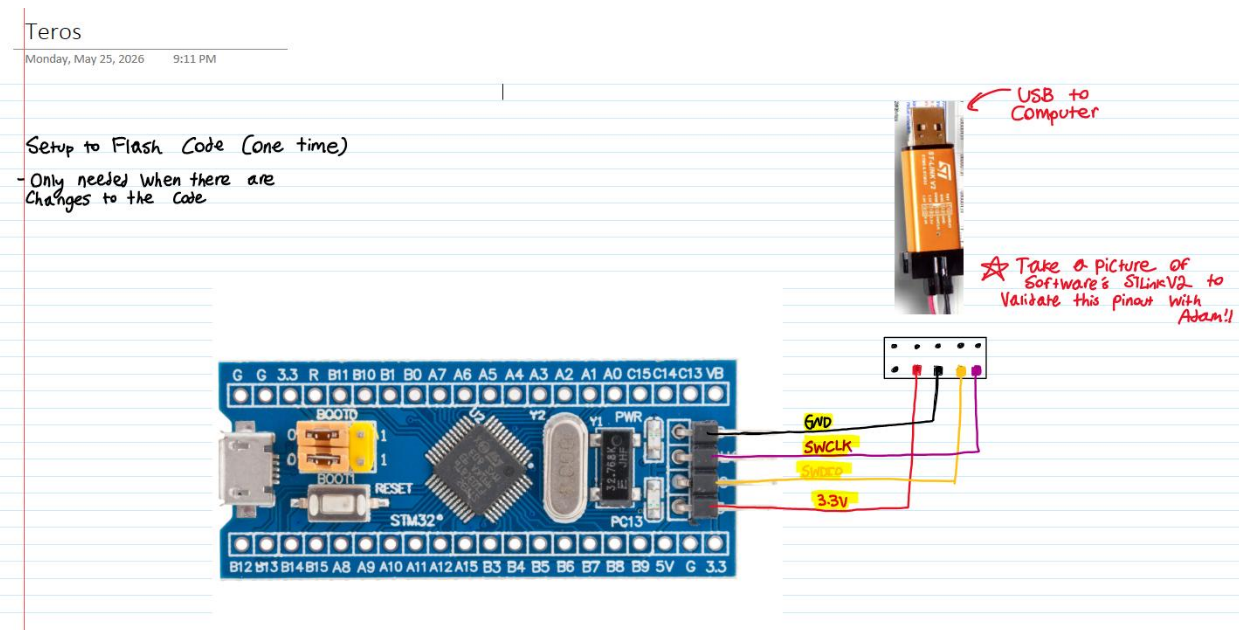
- Build with the checkmark in the top left corner. When complete, you should see the following message at the bottom of the screen. The code should build accordingly and without errors.



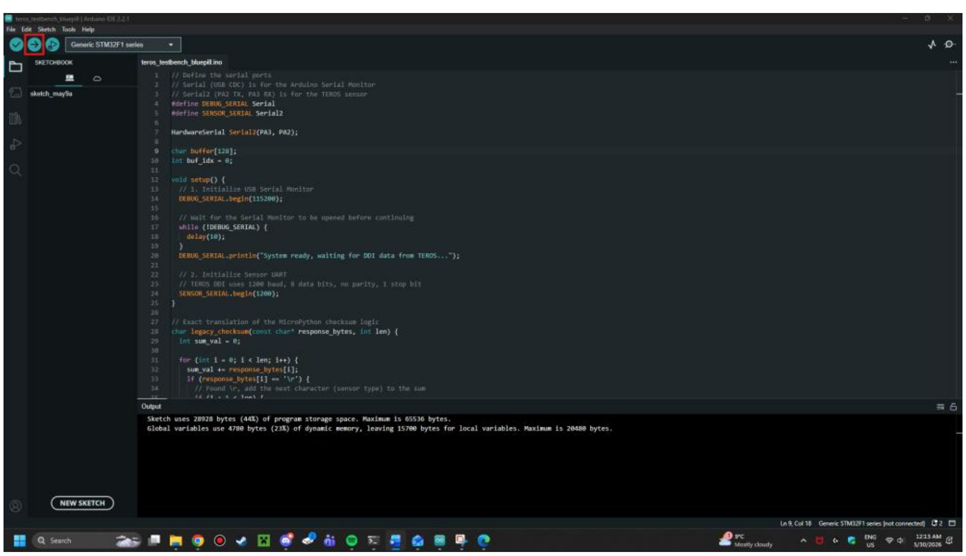
<u>From Blue Pill to STLink V2</u>

John has already assembled a hardware kit. You don't need to disassemble it, you should just be able to plug the STLink V2 into your computer over USB. If any wires are disconnected, the following pinout can be used to reconnect the STLinkv2 to the Blue Pill.

The STLink V2 should only be connected to the computer when you are flashing the firmware. At this time, the micro-USB cable for the Blue Pill should not be plugged in



- Connect the STLinkV2 to a USB Port on your computer
- Click the arrow to upload the code to the blue pill



- You should see a message in the bottom right corner or in the terminal stating that code flashing has been completed.
- **Unplug the STLink from the computer once this is done and the code has flashed successfully.**

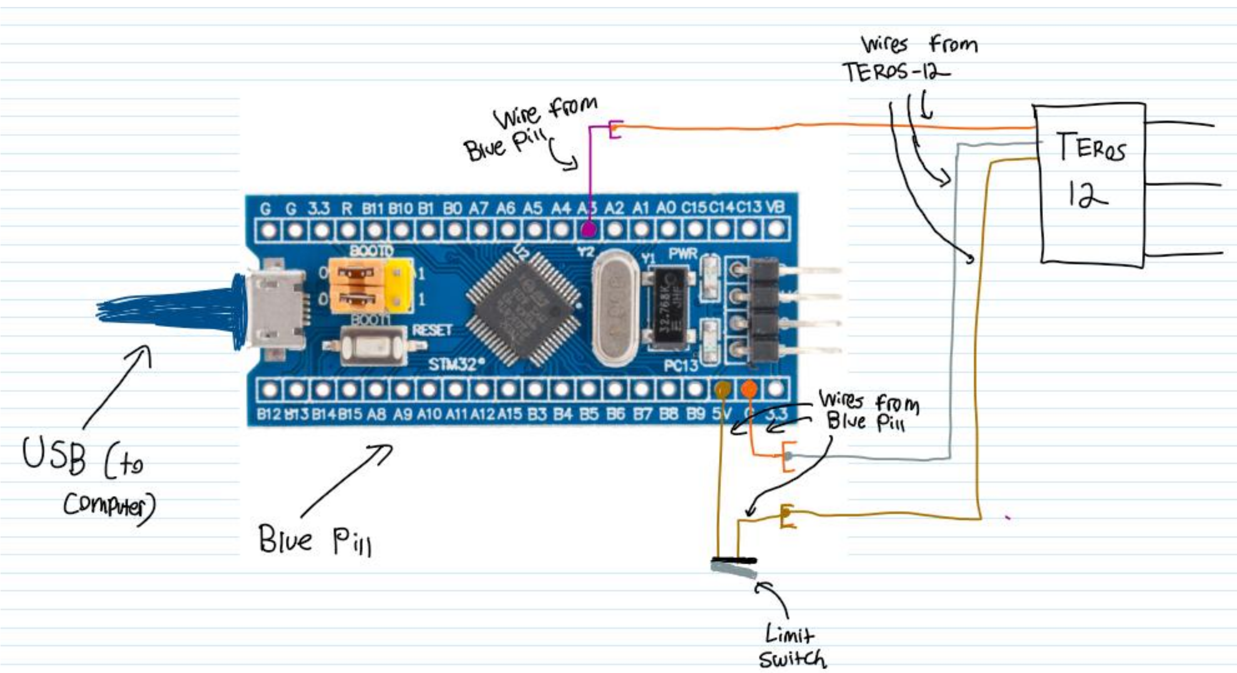
<u>Teros Testbench</u>

Below is a rough diagram of the TEROS testbench. John has already attached wires to different parts of the Blue Pill – however, please validate that they are in the correct place. **It is very likely that the limit switch is in the incorrect spot**. To connect the TEROS, plug the wires from the TEROS12 into the wires plugged into the following ports on the Blue Pill. **The top of the blue pill has all ports labelled on the left and right side.**

Teros Wire	Blue Pill Port	Notes
Brown	5V	**Connect to the opposite end of the limit switch**
Grey	GND	Any GND port will work – use the one that John already plugged a wire into
Orange	**A3**	

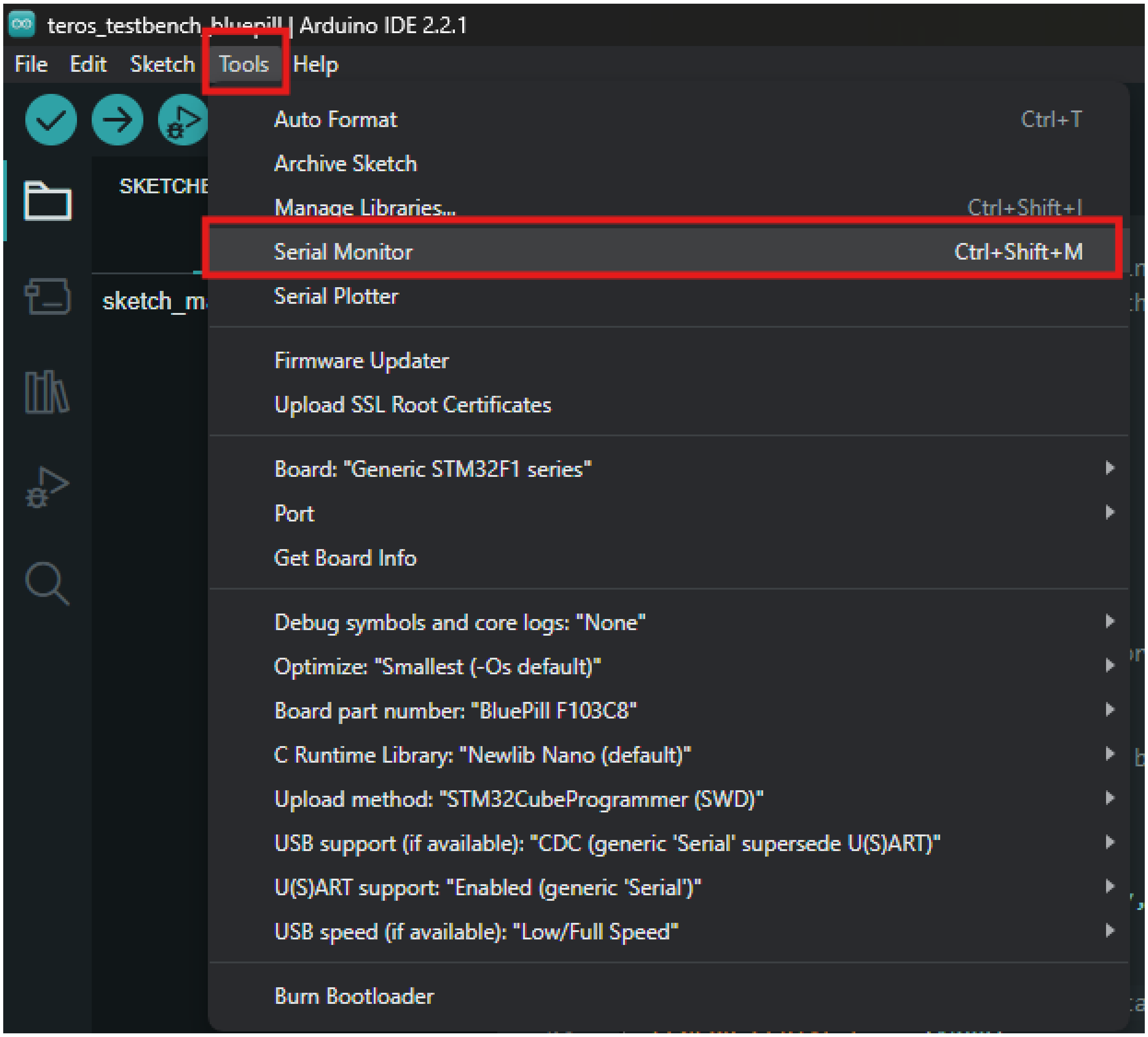
You can leave the STLink V2 plugged into the Blue Pill, but do not plug it to the computer after the code is flashed! Ensure it is unplugged before plugging the Blue Pill in!

Diagram of the setup (I've tried my best to map the wire colours, however they may vary)

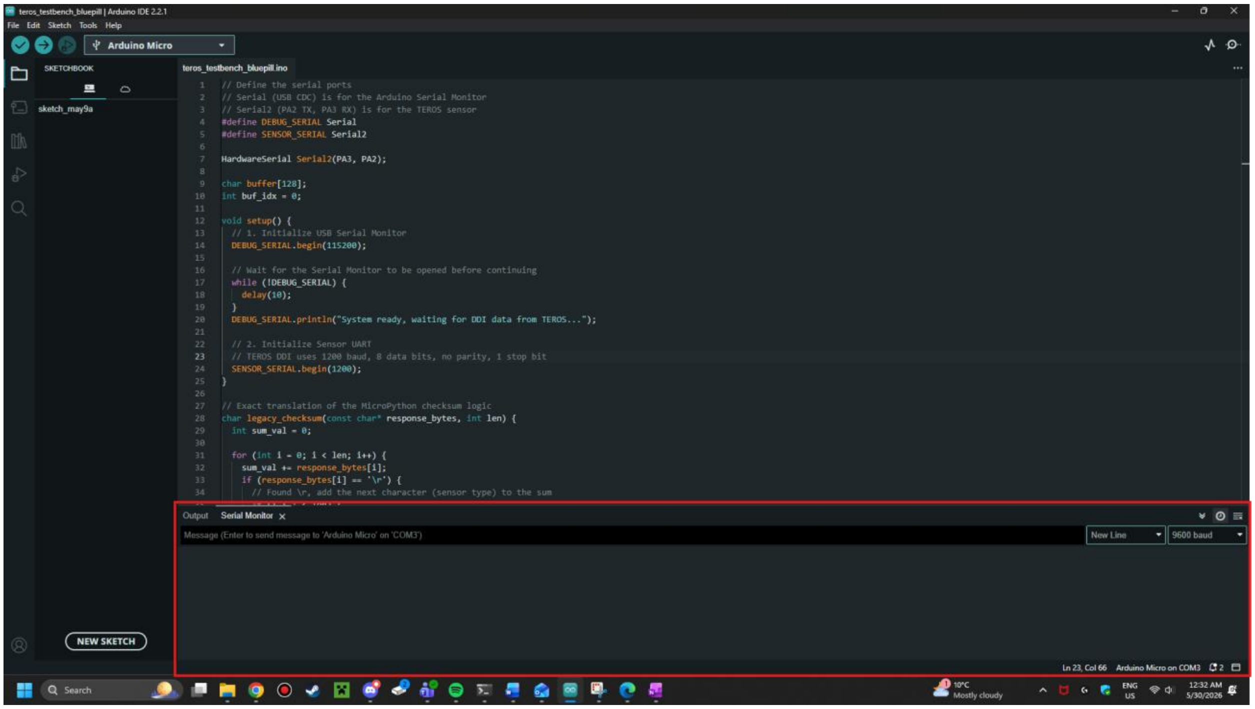


Viewing Data Output from the TEROS12

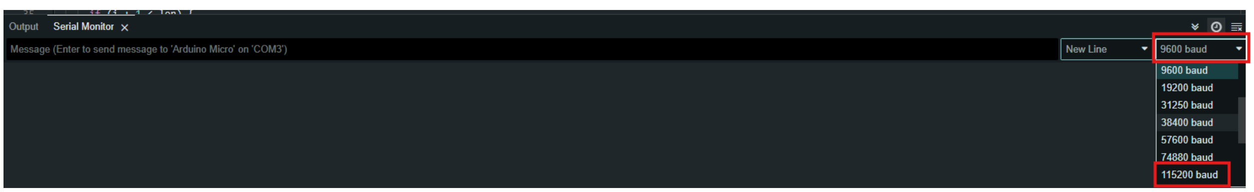
- Plug the Blue Pill into your computer via the attached USB cable (**not** using the STLink v2)
- In Arduino IDE, navigate to Tools > Serial Monitor



- The serial monitor will open at the bottom of the interface.



- In the top right corner of the Serial Montior, you will see something that says “9600 baud”. Open this and change the baud rate to **115200**.
    - You may have to scroll to find the correct baud rate.



Note: I had to connect a different microcontroller to get pictures of the serial monitor. Your interface and available baud rates may be slightly different

IMPORTANT: OBTAINING DATA FROM THE TEROS12

The TEROS12 does **not** send data automatically, and will only send data when power is cycled. We are emulating this through the limit switch, which essentially gives the TEROS12 power each time it is pressed!

To view information from the TEROS12, follow these steps:

- Ensure the Blue Pill is connected to the computer over USB (not via the STLink v2)
- Press and hold the limit switch until a new line shows in the Serial Monitor
- Release the limit switch.
- Repeat steps 2-3 for subsequent data values.

To disable the system, simply unplug the USB cable from your computer!

If you have any questions, do not hesitate to reach out to me on Teams either directly or via the sensors group chat!